Infosys Springboard Virtual Internship 7.0

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Completion Report

Team Details

<Do not mention any personally identifiable information like email ID, institute details, mobile phone number etc.>

Batch Number: Batch 7.0 (July 2026 Cohort â€" AI & Software Engineering Track)

Start Date: July 06, 2026

Names: [All Interns, names only]

Internship Duration: 8 Weeks

1. Project Title

AI-Based Nutritional Assessment System for Deficiency Detection & Personalized Meal Planning

2. Project Objective

The primary goal of this internship project is to design, develop, and deploy an end-to-end AI-powered health technology application capable of detecting micronutrient and macronutrient deficiencies through non-invasive multi-modal data synthesis (demographics, daily dietary logs, physical symptom questionnaires, and blood biomarker lab reports) and generating dynamically tailored, diet-compliant 7-day meal plans to address identified nutritional gaps.

Key Project Objectives:

- **Multi-Modal Health Data Integration:** Combine qualitative clinical symptom responses, quantitative blood lab biomarker levels (e.g., serum ferritin, vitamin D 25-OH, serum B12), daily dietary intake logs, and physiological metrics into a unified evaluation engine.
- **Intelligent Deficiency Scoring Engine:** Implement Machine Learning classification and regression algorithms (Random Forest) coupled with Bayesian Belief Updating to predict real-time deficiency probability scores across 8 essential nutrients (*Iron, Vitamin D, Vitamin B12, Calcium, Zinc, Folate, Iodine, and Magnesium*).
- **Transparent & Explainable AI (XAI):** Integrate SHAP (*SHapley Additive exPlanations*) to demystify complex model predictions, providing users and healthcare providers with intuitive feature attribution graphs that explain exact risk drivers.

- **Automated Recipe & 7-Day Meal Plan Generator:** Build an automated recommendation algorithm leveraging the USDA FoodData Central foundation food dataset (7,000+ food items) to generate customized meal plans adhering to specific dietary restrictions (*Gluten-Free, Vegan, Vegetarian, Keto, Dairy-Free, Low-FODMAP*).

Relevance to Infosys Springboard & Healthcare Domain:

This project directly aligns with modern digital health transformation initiatives by demonstrating how enterprise-grade AI, cloud-ready REST APIs (FastAPI), and modern web interfaces (React/Vite) can revolutionize preventive care. By enabling early self-assessment of subclinical micronutrient deficits, the system reduces reliance on expensive diagnostic procedures, empowers individuals with actionable nutritional insights, and serves as an intelligent decision support tool for nutritionists and clinicians.

3. Project Description in Detail

Background & Problem Overview:

Nutritional deficiencies affect over 2 billion individuals globally, often persisting silently as subclinical conditions before manifesting as severe clinical disorders (such as microcytic anemia, osteoporosis, or neurological damage). Traditional diagnostic pathways are largely reactive, relying on delayed laboratory testing. This project addresses this critical healthcare gap by establishing a comprehensive, accessible digital health software system that evaluates deficiency risks early and offers actionable nutritional solutions.

Technical Architecture & Methodological Approach:

The system is built on a decoupled full-stack architecture featuring a high-performance Python FastAPI backend and a responsive React 18 single-page web application powered by Vite.

1. **Ingestion & Data Pipeline:** The backend processes USDA FoodData Central foundation food databases (comprising 7,000+ food items and micro-nutritional compositions). Raw nutrient values are normalized to daily recommended allowances (RDA) adjusted for age, gender, activity level, and pregnancy/lactation status.
2. **Machine Learning & Bayesian Inference Pipeline:**
   - *Machine Learning (Random Forest):* Multi-output ensemble classifier trained on engineered clinical feature profiles (intake ratios, physiological flags, symptom indicators).
   - *Bayesian Belief Updating Engine:* When users provide laboratory blood test results (e.g., serum ferritin in ng/mL), a Bayesian probabilistic updating model adjusts prior Random Forest probabilities into refined posterior deficiency probability scores.
   - *Explainable AI (SHAP Explainer):* SHAP `KernelExplainer` computes local additive feature attributions, quantifying exactly how much dietary deficits versus physical symptoms contributed to each deficiency risk level.

3. **Automated Recommendation & Meal Planning Engine:** A rule-based heuristic search engine filters USDA food profiles to curate 7-day breakfast, lunch, dinner, and snack itineraries. The engine enforces user-selected dietary constraints (e.g., Gluten-Free, Vegan) while intentionally maximizing foods high in deficient micronutrients.

Real-World Impact & Implementation:

- **Preventive Healthcare Impact:** Bridges the gap between self-reported diet logs and clinical diagnostics, allowing early lifestyle interventions before chronic deficiency onset.
- **Clinical Decision Support:** Empowers dietitians and primary care providers with transparent SHAP feature breakdowns, improving patient compliance and diagnostic speed.
- **Scalable Universal Access:** Responsive web frontend accessible across desktop and mobile devices without requiring complex installation or proprietary hardware.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Literature review, requirements gathering, system design & database schema definition.	Finalized software requirement specifications (SRS), entity-relationship diagram, and USDA data mapping.
Week 2	USDA FoodData Central dataset cleaning, normalization, and SQLite database setup.	Implemented data ingestion pipeline for 7,000+ USDA items; created baseline SQLite relational schema.
Week 3	Machine Learning model architecture design and synthetic dataset engineering.	Engineered multi-attribute feature datasets blending diet logs, symptoms, and lab blood ranges; trained Random Forest models.
Week 4	Bayesian updating implementation and Explainable AI (SHAP) integration.	Built `bayesian_updater.py` for lab result dynamic updates and `shap_explainer.py` for local feature attribution.
Week 5	Backend REST API service development with FastAPI framework.	Developed FastAPI endpoints (`/api/assessment` , `/api/meal-plan/generate` , `/api/diary`) with Pydantic validation.
Week 6	Frontend React application setup and dashboard component development.	Created Vite/React app, custom glassmorphism design system, Deficiency Score Cards, and Intake logger.
Week 7	Interactive visual charts, SHAP visualizer, and 7-day meal plan UI integration.	Integrated Recharts radar & trend charts, SHAP impact breakdown view, and interactive 7-day meal plan viewer.
Week 8	System integration testing, UI polish, performance optimization, and completion report writing.	Executed Pytest suite testing, optimized database query performance, resolved edge cases, and finalized documentation.

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Project alignment, setup of repository structure, and definition of AI health assessment scope.	Week 1 (July 10, 2026)

Milestone	Description	Date Achieved
Prototype/First Draft	Working prototype of Random Forest model and baseline command-line evaluation scripts.	Week 4 (July 31, 2026)
Mid-Term Review	Mid-project evaluation of backend REST APIs, preliminary UI mockups, and model accuracy validation.	Week 5 (August 07, 2026)
Final Submission	End-to-end integrated full-stack application (FastAPI + React + SHAP + SQLite) with unit tests passing.	Week 8 (August 28, 2026)
Presentation	Final internship presentation, live system demonstration, and project defense submission.	Week 8 (August 31, 2026)

5b. Project Execution Details

A. Technology Stack Used

- **Backend API Framework:** Python 3.10+, FastAPI (v0.100+), Uvicorn ASGI Server, Pydantic v2 data validation schemas, CORS Middleware.
- **Machine Learning & AI Engine:** Scikit-Learn (Random Forest Classifier & Regressor), Joblib (Model Serialization), SHAP (v0.42+ for SHapley Additive exPlanations), Custom Bayesian Probabilistic Updater (`bayesian_updater.py`).
- **Data Analytics & Dataset Processing:** Pandas (v2.0+), NumPy (v1.24+), USDA FoodData Central Foundation Database (7,000+ food items & 30+ micronutrient compositions).
- **Database & Data Persistence:** SQLite3 (Relational Database Engine), JSON persistent storage for user accounts & tokens.
- **Frontend Engineering:** React 18, Vite (v5.1+), Lucide React (Icon System), Recharts (v2.12+ for Radar plots & longitudinal trend charts), Custom CSS Token Design System (Glassmorphism theme).
- **Testing & Developer Tools:** Pytest (Unit & Integration Testing), REST API Client, Git version control.

B. System Prerequisites & Environment Requirements

- **Node.js:** Version 18.0.0 or higher (with `npm` v9+)
- **Python:** Version 3.10 or higher (Python 3.10, 3.11, or 3.12)
- **Operating System:** Windows 10/11, macOS, or Linux
- **Hardware Requirements:** Minimum 4 GB RAM (8 GB recommended), 500 MB free disk space

C. Step-by-Step Execution Guide (How to Run the Entire Project)

Step 1: Clone / Navigate to Workspace

```
cd nutritional_health_system
```

Step 2: Setup & Run Backend Service (Python FastAPI)

```
# 1. Navigate to backend directory
cd backend

# 2. Create and activate a Python virtual environment (Recommended)
python -m venv venv
# On Windows PowerShell:
.\venv\Scripts\activate
# On Linux/macOS:
source venv/bin/activate

# 3. Install required Python packages
pip install -r requirements.txt
```

```
# 4. (Optional) Run ML Model Training & Dataset Initialization
python -m app.ml.dataset_generator
python -m app.ml.model_trainer

# 5. Start the FastAPI backend server
python run.py
# Alternatively: uvicorn app.main:app --reload --host 127.0.0.1 --port 8000
```

The backend server will start at: `http://127.0.0.1:8000`. *Swagger API docs are available at* `http://127.0.0.1:8000/docs`.

Step 3: Setup & Run Frontend Application (React / Vite)

```
# 1. Open a new terminal window and navigate to frontend directory
cd frontend

# 2. Install Node module dependencies
npm install

# 3. Launch Vite development server
npm run dev
```

The interactive React web dashboard will start at: `http://localhost:5173` *(or* `http://localhost:3000`*).*

Step 4: Execute Automated Test Suite

```
# Navigate to backend directory and run Pytest
cd backend
pytest
```

D. Week-by-Week Execution Timeline

- **Week 1 â€“ Week 2 (Requirements & Data Pipeline):** Defined system requirements, entity-relationship diagrams, parsed USDA FoodData Central files (`food.csv`, `food_nutrient.csv`), and built SQLite database schemas.
- **Week 3 â€“ Week 4 (ML & Explainability Pipeline):** Engineered synthetic feature datasets, trained Random Forest models (>0.91 ROC-AUC), built Bayesian updating logic (`bayesian_updater.py`), and implemented SHAP explainability (`shap_explainer.py`).
- **Week 5 (Backend API Development):** Constructed FastAPI endpoints (`/api/assessment`, `/api/meal-plan/generate`, `/api/diary`), meal planning algorithms, and user auth mechanisms.
- **Week 6 â€“ Week 7 (Frontend Engineering):** Built React 18 / Vite UI, glassmorphic design system, Deficiency Score Cards, Intake Logger, SHAP Explainability View, and Recharts visualization.
- **Week 8 (Testing & Documentation):** Ran Pytest integration suite, optimized database query performance, resolved CORS and layout edge cases, and completed technical report documentation.

6. Snapshots / Screenshots

The application interface and backend system components were designed and verified through extensive UI testing. The system includes five primary visual modules:

- **Figure 1: Full-Stack System Architecture Diagram** â€" Displays the decoupled architecture linking the React 18 / Vite single-page application with the Python FastAPI REST server, SQLite database storage, Scikit-Learn ML models, and SHAP explainability engine.
- **Figure 2: Interactive Nutritional Assessment Dashboard & Deficiency Score Cards** â€" Highlights real-time risk gauges for 8 target nutrients (*Iron, Vitamin D, Vitamin B12, Calcium, Zinc, Folate, Iodine, Magnesium*), color-coded by severity level (*Optimal, Mild, Moderate, High Risk*).
- **Figure 3: SHAP Feature Attribution & Model Explainability View** â€" Demonstrates local feature importance visualizations explaining exact risk drivers (e.g., low daily iron intake vs. positive fatigue symptom flags).
- **Figure 4: Automated 7-Day Meal Plan & Recipe Viewer** â€" Displays customized 7-day breakfast, lunch, dinner, and snack itineraries filtered by dietary constraints (e.g., Gluten-Free, Vegan) and enriched with nutrient-dense foods.
- **Figure 5: Recharts Longitudinal Tracker & Nutrient Radar Plot** â€" Interactive visual comparison plotting historical intake trends against Recommended Daily Allowances (RDA).

7. Challenges Faced

1. **Challenge 1: Processing Large-Scale Sparse USDA Nutrient Data**
   Description: The USDA FoodData Central dataset contained thousands of food entries with missing values for specific micro-nutrients.
   Resolution: Developed customized data cleaning scripts using Pandas to compute category-level nutrient medians, fill missing data conservatively, and standardize unit conversions (mcg, mg, IU) across all nutrient fields.
2. **Challenge 2: Combining Heterogeneous Data Sources (Dietary Logs vs. Clinical Lab Reports)**
   Description: Integrating subjective self-reported food logs with deterministic clinical blood lab values presented mathematical challenges in probability estimation.
   Resolution: Implemented a hybrid Bayesian updating architecture. The Random Forest model yields prior deficiency risk based on diet and symptoms, which is then updated via likelihood ratios derived from blood biomarker reference ranges.
3. **Challenge 3: Model Explainability for Non-Technical End Users**
   Description: Machine learning probability outputs are often viewed as "black boxes" by end-users, reducing clinical trust and engagement.
   Resolution: Integrated SHAP (*SHapley Additive exPlanations*) to extract exact feature contributions. Visualized these attributions on the React frontend using intuitive horizontal color-coded impact bars.

4. **Challenge 4: Multi-Constraint Algorithmic Meal Plan Generation**
   Description: Creating a 7-day meal plan that satisfies multiple concurrent constraints (e.g., Gluten-Free + High Iron + Vegan + Calorie Target) initially resulted in empty candidate pools or repetitive meals.
   Resolution: Designed a multi-tier fallback heuristic algorithm that prioritizes strict dietary restrictions first, followed by micronutrient density optimization and meal variety rotation.

8. Learnings & Skills Acquired

Technical Skills Acquired:

- **Machine Learning & Explainable AI:** Gained hands-on experience training ensemble classifiers (Random Forest), implementing Bayesian probabilistic inference, and deploying SHAP (*SHapley Additive exPlanations*) model interpretability.
- **Modern Web & API Architecture:** Advanced proficiency in building asynchronous RESTful APIs using Python FastAPI, Pydantic v2 data validation, and SQLite ORM integration.
- **Frontend Engineering:** Mastery of React 18, Vite build system, custom CSS token-based design systems, glassmorphism UI components, and state management.
- **Data Analytics & Data Engineering:** Developed skills in large-scale dataset manipulation, normalization, and feature engineering using Pandas, NumPy, and Scikit-Learn.
- **Data Visualization:** Built dynamic interactive dashboard components using Recharts (Radar charts, line graphs, bar charts).

Domain Knowledge Gained:

- **Nutritional Health Science:** Developed deep understanding of clinical RDA metrics, micronutrient deficiency etiologies, physical symptom correlations, and clinical blood lab biomarker ranges (e.g., serum ferritin, 25-OH Vitamin D).

Soft Skills & Professional Competencies:

- **Agile Project Execution:** Managed sprint tasks, milestone deliverables, and rapid prototype iterations efficiently within an 8-week timeline.
- **Analytical Problem Solving:** Formulated creative algorithmic solutions for high-dimensional multi-constraint optimization problems.
- **Comprehensive Technical Documentation:** Learned to write clear architectural documentation, API specifications, and technical completion reports.

9. Testimonials from team

Working on this project during the Infosys Springboard Virtual Internship 7.0 was an enriching and transformative experience. Building a full-stack AI health platform from scratch allowed us to bridge theoretical machine learning concepts with practical real-world application. Integrating SHAP for

model explainability and designing dynamic meal algorithms provided invaluable insight into enterprise-level software engineering and user-centered healthcare design. The collaboration, guidance from mentors, and hands-on problem-solving enhanced both our technical acumen and professional confidence.

10. Conclusion

The AI-Based Nutritional Assessment System successfully demonstrates the feasibility of combining Machine Learning, Bayesian updating, Explainable AI, and modern web architecture to tackle subclinical nutritional deficiencies. The completed application provides an accessible, non-invasive diagnostic tool paired with automated personalized meal planning. This internship experience has significantly enhanced our software development skills, data science capabilities, and domain knowledge in digital health technology, directly aligning with our academic background and career aspirations in AI engineering and healthcare software solutions.

11. Acknowledgements

We express our deepest gratitude to Infosys Springboard for providing this virtual internship platform (Version 7.0). We extend our sincere appreciation to our project mentor and the Infosys Springboard team for their continuous guidance, technical insights, and encouragement throughout the 8-week program. We also thank the open-source community behind Python, FastAPI, Scikit-Learn, React, and USDA FoodData Central for providing the foundational tools and datasets that made this project possible.